

Assignment 3: Creative Core Polish

Overview

In this assignment, you will apply what you learned in the Creative Core pathway to polish your Knockout Arena project. You must add at least **one enhancement from each** of the six sections below. You will also submit a short written report documenting what you added and where in the project it can be found.

This is a lighter assignment. The goal is intentional application of Creative Core skills, not a large new feature set. Quality and intentionality matter more than quantity.

Section 1: Shaders and Materials

Add at least one shader or material enhancement. This could be a custom material using URP/Lit shader properties (metallic, smoothness, emission), a Shader Graph shader for a gameplay effect (damage flash, dissolve, pulsing glow), or runtime material changes via script (e.g., tinting enemies when low HP, shifting arena color per wave).

Section 2: Lighting

Add at least one meaningful lighting enhancement. This could be configuring the Directional Light and shadows to set a mood, adding Point or Spot Lights to highlight gameplay elements (powerup spawn, arena edges), changing light properties dynamically in code (flickering, intensity that scales with wave number), or baking a lightmap for static geometry.

Section 3: Animation

Add at least one animation improvement beyond what the original Knockout Arena worksheet required. This could be a well-structured Animator Controller with proper state transitions and conditions, keyframe animations via the Animation window (arena bobbing, powerup floating, camera shake), animated UI elements, or animation control from code using `SetBool()`, `SetTrigger()`, or `SetFloat()` tied to gameplay events.

Section 4: VFX

Add at least one new Particle System effect beyond the single particle effect the original worksheet required. This should be for a different game event -- for example, a burst on enemy knockout, environmental particles (dust, embers, rain), a powerup collection ring, or a trail on the player while moving. Control all new effects from code using `.Play()` and/or `.Stop()`.

Section 5: Cameras

Add at least one camera enhancement beyond the basic focal-point orbit. This could be a scripted camera shake on impact, a smooth zoom that pulls out as waves increase, a dynamic FOV change while powered up, or a brief cinematic camera move at the start of each wave.

Section 6: Post-Processing

Add a URP Volume with at least one post-processing effect. This could be Bloom (to make emissive materials and lights glow), Vignette (to darken screen edges for mood), Color Grading (to shift the overall palette), or a gameplay-driven override (e.g., increasing Chromatic Aberration or Vignette intensity when powered up). Use a Global Volume for scene-wide effects or a Local Volume tied to a specific area.

Section 7: Written Report

Submit a document (PDF or README.md) that lists each of the six sections above and, for each one, describes in 2-3 sentences: what you implemented, which GameObject(s) or script(s) it involves, and how to see it in action when running the project. Include screenshots where helpful.

Submission

Submit a .zip file or GitHub repository link containing your complete Unity project and your written report. Make sure the project opens and runs without errors in Unity 6.x.

Resources

- [Creative Core: Shaders and Materials](#)
- [Creative Core: Lighting](#)
- [Creative Core: Animation](#)
- [Creative Core: VFX](#)
- [Creative Core: Cameras](#)
- [Creative Core: Post-Processing](#)
- [Unity 6 Documentation](#)